EXTENSION BY ZERO ALONG AN ETALE MORPHISM COMMUTES WITH BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: both adjunctions and the identification of the right adjoints are displayed, and the conclusion follows by uniqueness of adjoints. The Xy-pic square of the counted statement is delivered as written.

Editorial note. The label lemma-shriek-base-change occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The single labelled reference occurring inside the counted spans points to a chapter of the Project not staged in this card's source tree; it is carried in assembly.external_references as a non-load-bearing navigation pointer recorded at the invoking line.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Extension by zero j_! along an etale morphism j : V -> X. Same chapter, the functor j_! on abelian sheaves and on sheaves of modules, left adjoint to the restriction j^{-1}. Supplies the functors appearing on both sides of the asserted identity.
- Adjunction identifying j_! as a left adjoint. Same chapter, the adjunction between j_! and j^{-1}, and between f^{-1} and f_*. Supplies the argument that two functors with the same right adjoint are equal.
- Commutation of pushforward with etale localization. Same chapter, the identification f'_*(j')^{-1} = j^{-1}f_* coming from the construction of pullback and pushforward. Supplies the equality of right adjoints which forces the equality of the left adjoints.

Source reference. Original lemma 04J0: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11512–11512. Cross-chapter navigation pointer invoked at proof lines 11510-11512, where the author notes that the statement holds in the general setting of a morphism of sites. The target chapter is not staged in this card's source tree, so the pointer records the invoking line; it only resolves the reference for the reader, and no bibliography citation is involved.

Lemma 0.1 (Extension by zero commutes with base change)**.** *Let $f : Y \to X$ be a morphism of schemes. Let $j : V \to X$ be an étale morphism. Consider the fibre*

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product

$$\begin{array}{ccc} V' = Y \times_X V & \xrightarrow{j'} & Y \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ V & \xrightarrow{j} & X \end{array}$$

Then we have $j'_! f'^{-1} = f^{-1} j_!$ on abelian sheaves and on sheaves of modules.

Proof. This is true because $j'_! f'^{-1}$ is left adjoint to $f'_*(j')^{-1}$ and $f^{-1} j_!$ is left adjoint to $j^{-1} f_*$. Further $f'_*(j')^{-1} = j^{-1} f_*$ because f_* commutes with étale localization (by construction). In fact, the lemma holds very generally in the setting of a morphism of sites, see Modules on Sites, Lemma 04J0. □